ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{23}{24}$	Use the LCM of the denominators.
2	**(d)** 60	Difference of the fractions, then divide.
3	**(b)** 40	Fraction removed = difference.
4	**(a)** 8	Divide the length by the piece length.
5	**(c)** 5	Find the number first.
6	**(d)** $\frac{7}{10}$	Compare with a common denominator.
7	**(a)** $2\frac{3}{4}$	Subtract.
8	**(b)** 24	LCM.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Multiplying 3 and 4 by 2 gives 6 and 8.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true. R is just another way of writing the number and does not explain why it is improper.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number $= 18 \div \frac{3}{4} = 24$ (1); $\frac{1}{3}$ of the number $= 8$ (1).	1+1
12	Required $= \frac{5}{6} - \frac{2}{7}$ (1); $= \frac{23}{42}$ (1).	1+1
13	One way $= \frac{23}{20}$ km (1); Total $= 2 \times = \frac{23}{10}$ km (1).	1+1
14	Number $= 14 \div \frac{2}{5} = 35$ (1); $\frac{3}{7}$ of the number $= 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Girls $= \frac{5}{8} \times 480 = 300$ (1); Chess girls $= \frac{2}{5} \times 300 = 120$ (1); Boys $= 480 - 300 = 180$ (1).	1+1+1
16	$\frac{?}{12} = \frac{5}{6} - \frac{1}{4}$ (1); $= \frac{7}{12} = \frac{7}{12}$ (1); Missing numerator $= 7$ (1).	1+1+1
17	Together $= \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$ (1); Remaining $= 1 - \frac{5}{12} = \frac{7}{12}$ (1); State the answer (1).	1+1+1
	OR: Girls $= \frac{4}{9} \times 360 = 160$ (1); Chess girls $= \frac{3}{8} \times 160 = 60$ (1); Boys $= 360 - 160 = 200$ (1).	1+1+1
18	$\frac{?}{20} = \frac{3}{4} - \frac{1}{5}$ (1); $= \frac{11}{20} = \frac{11}{20}$ (1); Missing numerator $= 11$ (1).	1+1+1
	OR: Together $= \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$ (1); Remaining $= 1 - \frac{8}{15} = \frac{7}{15}$ (1); State the answer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$ (1). (ii) $1 - \frac{5}{8} = \frac{3}{8}$ (1). (iii) Kavita (3 slices against 2) (1). (iv) 3 slices are left (1); $\frac{3}{8} = \frac{6}{16}$ (1).	1+1+1+2
20	After day 1: $\frac{3}{4}$ left (1); Day 2 sells $\frac{1}{2}$ of total (1); Left fraction $\frac{1}{4}$ (1); Total $= 12 \div \frac{1}{4} = 48$ m (2).	1+1+1+2
	OR: Sum of whole parts $= 4$ (1); Sum of fractions $= \frac{7}{8}$ (1); Total $= 4\frac{7}{8}$ (1); After subtracting: $4\frac{3}{8}$ (2).	1+1+1+2

Mistakes to avoid:

- Adding the denominators as well: $\frac{1}{5} + \frac{2}{5} = \frac{3}{7}$ is wrong; the answer is $\frac{3}{5}$.
- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2 \times 2 + 1}{2} = \frac{5}{2}$.
- Comparing fractions by denominators only: a bigger denominator with the same numerator makes the fraction *smaller*.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 48	Fraction removed = difference.
2	**(c)** 15	Divide the length by the piece length.
3	**(c)** $10\frac{1}{2}$	Find the number first.
4	**(b)** $\frac{11}{15}$	Compare with a common denominator.
5	**(b)** $1\frac{5}{6}$	Subtract.
6	**(b)** 18	LCM.
7	**(d)** 1	Use the LCM of the denominators.
8	**(d)** 24	Difference of the fractions, then divide.
9	**(c)** A is true but R is false.	A is true; the denominator stays the same, so R is false.
10	**(d)** A is false but R is true.	$\frac{3}{8} = 0.375 < 0.5 = \frac{1}{2}$, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Required =\frac{11}{12}−\frac{3}{8} (1); = $\frac{13}{24}$ (1).	1+1
12	One way = $\frac{13}{12}$ km (1); Total = 2× = $\frac{13}{6}$ km (1).	1+1
13	Number = 25÷\frac{5}{6}= 30 (1); $\frac{2}{5}$ of the number = 12 (1).	1+1
14	Required =\frac{7}{9}−\frac{1}{6} (1); = $\frac{11}{18}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Girls =\frac{3}{7}×420 = 180 (1); Chess girls =\frac{5}{9}×180 = 100 (1); Boys = 420 − 180 = 240 (1).	1+1+1
16	$\frac{?}{24}$ =\frac{7}{8}−\frac{1}{6} (1); = $\frac{17}{24}$ =\frac{17}{24} (1); Missing numerator = 17 (1).	1+1+1
17	Together =\frac{1}{5}+\frac{1}{4}= $\frac{9}{20}$ (1); Remaining = $1 - \frac{9}{20} = \frac{11}{20}$ (1); State the answer (1).	1+1+1
	OR: Girls =\frac{7}{12}×240 = 140 (1); Chess girls =\frac{3}{7}×140 = 60 (1); Boys = 240 − 140 = 100 (1).	1+1+1
18	$\frac{?}{30}$ =\frac{5}{6}−\frac{1}{3} (1); = $\frac{1}{2}$ =\frac{15}{30} (1); Missing numerator = 15 (1).	1+1+1
	OR: Together =\frac{1}{6}+\frac{1}{8}= $\frac{7}{24}$ (1); Remaining = $1 - \frac{7}{24} = \frac{17}{24}$ (1); State the answer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{5}{2}$ (1). (ii) $2\frac{1}{2} + 1\frac{3}{4} = \frac{10}{4} + \frac{7}{4} = \frac{17}{4} = 4\frac{1}{4}$ litres (1). (iii) Morning, as $2\frac{1}{2} > 1\frac{3}{4}$ (1). (iv) $4\frac{1}{4} - 3\frac{1}{2} = \frac{17}{4} - \frac{14}{4} = \frac{3}{4}$ litre (2).	1+1+1+2
20	After day 1: \frac{1}{2} left (1); Day 2 sells $\frac{1}{10}$ of total (1); Left fraction $\frac{2}{5}$ (1); Total = $16 \div \frac{2}{5} = 40$ m (2).	1+1+1+2
	OR: Sum of whole parts = 4 (1); Sum of fractions = $\frac{5}{3}$ (1); Total = $5\frac{2}{3}$ (1); After subtracting: $5\frac{1}{12}$ (2).	1+1+1+2

Mistakes to avoid:

- Adding the denominators as well: $\frac{1}{5} + \frac{2}{5} = \frac{3}{7}$ is wrong; the answer is $\frac{3}{5}$.
- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2\times2+1}{2} = \frac{5}{2}$.
- Comparing fractions by denominators only: a bigger denominator with the same numerator makes the fraction *smaller*.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Fractions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 24	Find the number first.
2	**(d)** $\frac{7}{12}$	Compare with a common denominator.
3	**(d)** $3\frac{9}{10}$	Subtract.
4	**(a)** 30	LCM.
5	**(b)** $\frac{3}{4}$	Use the LCM of the denominators.
6	**(d)** 40	Difference of the fractions, then divide.
7	**(a)** 42	Fraction removed = difference.
8	**(b)** 12	Divide the length by the piece length.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{3}{6}+\frac{2}{6}=\frac{5}{6}$ using the common denominator 6.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	12 / 3 = 4 and 4 x 2 = 8. Both true, but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One way $=\frac{47}{40}$ km (1); Total $=2\times=\frac{47}{20}$ km (1).	1+1
12	Number $=16\div$\frac{4}{9}$=36$ (1); $\frac{3}{4}$ of the number $=27$ (1).	1+1
13	Required =\frac{9}{10}−\frac{4}{15} (1); $=\frac{19}{30}$ (1).	1+1
14	One way $=\frac{11}{9}$ km (1); Total $=2\times=\frac{22}{9}$ km (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Girls =\frac{5}{6}$\times 540=450$ (1); Chess girls =\frac{2}{9}$\times 450=100$ (1); Boys $=540-450=90$ (1).	1+1+1
16	$\frac{?}{18}$ =\frac{5}{6}−\frac{1}{9} (1); $=\frac{13}{18}$ =\frac{13}{18} (1); Missing numerator $=13$ (1).	1+1+1
17	Together =\frac{2}{9}+\frac{1}{3}$=\frac{5}{9}$ (1); Remaining $=1-\frac{5}{9}=\frac{4}{9}$ (1); State the answer (1).	1+1+1
	OR: Girls =\frac{3}{5}$\times 300=180$ (1); Chess girls =\frac{5}{6}$\times 180=150$ (1); Boys $=300-180=120$ (1).	1+1+1
18	$\frac{?}{16}$ =\frac{7}{8}−\frac{1}{4} (1); $=\frac{5}{8}$ =\frac{10}{16} (1); Missing numerator $=10$ (1).	1+1+1
	OR: Together =\frac{1}{7}+\frac{1}{4}$=\frac{11}{28}$ (1); Remaining $=1-\frac{11}{28}=\frac{17}{28}$ (1); State the answer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{3}{5}\times 40=24$ (1). (ii) $\frac{1}{4}\times 40=10$ (1). (iii) $40-24-10=6$ (1). (iv) $\frac{6}{40}=\frac{3}{20}$ (the check: $1-\frac{3}{5}-\frac{1}{4}=\frac{3}{20}$) (2).	1+1+1+2
20	After day 1: \frac{4}{5} left (1); Day 2 sells $\frac{3}{10}$ of total (1); Left fraction $\frac{1}{2}$ (1); Total $=25\div\frac{1}{2}=50$ m (2).	1+1+1+2
	OR: Sum of whole parts $=7$ (1); Sum of fractions $=\frac{3}{4}$ (1); Total $=7\frac{3}{4}$ (1); After subtracting: $6\frac{5}{6}$ (2).	1+1+1+2

Mistakes to avoid:

- Adding the denominators as well: $\frac{1}{5}+\frac{2}{5}=\frac{3}{7}$ is wrong; the answer is $\frac{3}{5}$.
- Adding numerators and denominators of unlike fractions directly, e.g. $\frac{1}{2}+\frac{1}{3}=\frac{2}{5}$.
- Changing only the numerator or only the denominator when making an equivalent fraction.
- Converting $2\frac{1}{2}$ to $\frac{3}{2}$ instead of $\frac{2\times2+1}{2}=\frac{5}{2}$.
- Comparing fractions by denominators only: a bigger denominator with the same numerator makes the fraction *smaller*.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.